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Reaching a Nonlinear Consensus: Polynomial Stochastic Operators

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Abstract: We provide a general nonlinear protocol for a structured time-varying and synchronous multi-agent system. We present an opinion sharing dynamics of the multi-agent system as a trajectory of a polynomial stochastic operator associated with a multidimensional stochastic hypermatrix. We show that the multi-agent system eventually reaches to a consensus if either one of the following two conditions is satisfied: (i) every member of the group people has a positive subjective opinion on the given task after some revision steps or (ii) all entries of a multidimensional stochastic hypermatrix are positive. Numerical results are also presented.

Keywords: Multidimension stochastic hypermatrix, nonlinear consensus, polynomial operator.

1. INTRODUCTION

Historically, an idea of reaching consensus through repeated averaging was introduced by DeGroot [1] for a structured time-invariant and synchronous environment. Since that time, the consensus which is the most ubiquitous phenomenon of multi-agent systems becomes popular in various scientific communities, such as biology, physics, control engineering and social science [2-20]. The dynamics of opinion sharing, competing, and the emergence of consensus have became an active topic of the recent research in statistical and nonlinear physics [21]. For example, there are many models which address to how consensus can be achieved in the evolution of two competing opinion in a population, which include the voter model [22], the majority rule model [23,24], and the social impact model [25]. Due to the high relevance of complex networks [26] to social and natural systems, opinion dynamics have also been incorporated on networks [27-40] such as regular lattices, random graphs [41], small-world networks [42] and scale-free networks [43]. Phase transitions in opinion dynamics were observed in [27,28,31,33,34] and the emergence of global consensus was investigated in [22, 24] where all agents share the same opinion. It has also been found that both the network structures [30] and the opinion updating strategies [32,35,36] can affect the time for reaching the final consensus. To our best knowledge, most research papers are concerned with the consensus problem under linear protocols. However, many systems, such as the well-known Kuramoto oscillator, exhibit nonlinear, locally passive dynamics as discussed in [44]. The consensus problem in some nonlinear protocols were studied in [5-20]. In this paper, we are aiming to provide some nonlinear protocols of multi-agent systems and to study the consensus problem for the provided nonlinear protocols. For reasons of self-exposition, it is convenient to first provide a linear protocol for an estimate-modification process of a structured time-invariant and synchronous environment which was presented in [1,45]. We consider a group of m individuals $I = \{1, 2, \cdots, m\}$ each of whom can specify his or her own subjective probability distribution for some given task. Suppose the m individuals must act together as a team or committee. Let $x_i^{(0)}$ denote the subjective distribution that the individual i assigns to the given task. The subjective distributions, $x^{(0)} = (x_1^{(0)}, \cdots, x_m^{(0)})^T$, will be based on the different backgrounds and different levels of expertise of the members of the group. It is assumed that if the individual i is informed of the distributions of each of the other members of the group, he/she might wish to revise his/her subjective distribution to accommodate this information. In the DeGroot's model [1], it was assumed that when the individual i makes this revision, his/her revised distribution is a linear combination of the distributions $x_1^{(0)}, \cdots, x_m^{(0)}$. Let p_{ij} denote the weight that the individual i assigns to $x_j^{(0)}$ when he/she makes this revision. It was assumed that $p_{ij} \geq 0$ and $\sum_{j=1}^{m} p_{ij} = 1$. So, after being informed of the subjective distributions of the other members of the group, the individual i revises his/her own subjective distribution from $x_i^{(0)}$ to $x_i^{(1)} = \sum_{j=1}^{m} p_{ij} x_j^{(0)}$. It is clear that $\mathbb{P} = (p_{ij})_{i,j=1}^{m}$ is a row-stochastic matrix since all elements are non-negative and the row-sums are equal to one.

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Let $x^{(0)} = (x_1^{(0)}, \cdots, x_m^{(0)})^T$, $x^{(1)} = (x_1^{(1)}, \cdots, x_m^{(1)})^T$ be vectors. Then the vector of revised subjective distributions can be written as $x^{(1)} = \mathbb{P}x^{(0)}$. The critical step in this process is that the above revision is iterated. Let $x^{(n)} = (x_1^{(n)}, \cdots, x_m^{(n)})^T$ denote the subjective distribution of the members of the group after *n* step revisions. Then $x^{(n)} = \mathbb{P}x^{(n-1)} = \mathbb{P}^n x^{(0)}$. In [1], DeGroot states that a consensus is reached if and only if all *m* components of $x^{(n)}$ converge to the same limit as $m \to \infty$.

In this paper, our main assumption is that the subjective distribution $x^{(n)} = (x_1^{(n)}, \cdots, x_m^{(n)})^T$ of the members is probabilistic in every step, i.e., $x_i \geq 0$ and $\sum_{i=1}^{m} x_i =$ 1 for any $i = \overline{1,m}$ and $n \in \mathbb{N}$. Let S^{m-1} be an $(m-1)$–dimensional simplex, where

$$S^{m-1} = \left\{ x \in \mathbb{R}^m : \sum_{i=1}^{m} x_i = 1,\ x_i \geq 0,\ i = \overline{1,m} \right\}.$$

It is worth mentioning that, for DeGroot's model, one has that $x^{(n)} \in S^{m-1}$ for any $n \in \mathbb{N}$ if and only if $\sum_{j=1}^{m} p_{ij} = \sum_{i=1}^{m} p_{ij} = 1$ and $p_{ij} \geq 0$ i.e., $\mathbb{P} = (p_{ij})_{i,j=1}^{m}$ is the doubly stochastic matrix. Consequently, we may conclude that a trajectory $\{x^{(n)}\}$ of the doubly stochastic matrix $\mathbb{P}$ presents the DeGroot model of the structured time-invariant synchronous environment with the probabilistic subjective distribution for some given task. In [46], Chatterjee and Seneta consider a generalization of DeGroot's model in which the individuals can change their weights p_{ij} at each iteration. More precisely, let $\{\mathbb{P}_n\}$ be a sequence of doubly stochastic matrices (a non-homogeneous Markov chain) and $x^{(0)} \in S^{m-1}$. A sequence $x^{(n+1)} = \mathbb{P}_n x^{(n)}$ presents the Chatterjee-Seneta model of the structured time-varying synchronous environment with the probabilistic subjective distribution for some given task.

In this paper, we shall consider a nonlinear model for the estimate modification process of the structured time-varying synchronous environment which generalizes both the DeGroot and Chatterjee-Seneta models.

In general, we suppose that doubly stochastic matrices (in the Chatterjee-Seneta model) depend on subjective distributions $x^{(n)}$ in every step, i.e., entries of doubly stochastic matrices are not constants but functions of $x^{(n)}$,

$$\mathbb{P}_{x^{(n)}} = \left(p_{ij}(x^{(n)}) \right)_{i,j=1}^{m}. \tag{1}$$

A general model of the structured time-varying synchronous environment with probabilistic subjective distributions is defined as follows:

$$x^{(n+1)} = \mathbb{P}_{x^{(n)}} x^{(n)}, \tag{2}$$

where $\mathbb{P}_{x^{(n)}}$ is a doubly stochastic matrix defined by (1). The choese of $\mathbb{P}_{x^{(n)}}$, we may get different models of multi-agent systems. For instance, if $\mathbb{P}_{x^{(n)}} = \mathbb{P}_0$ (the matrices are free of $x^{(n)}$ and *n*) then we get the DeGroot model. If $\mathbb{P}_{x^{(n)}} = \mathbb{P}_n$ (the matrices are free of $x^{(n)}$ but depended on *n*) then we get the Chatterjee-Seneta model. We shall study a consensus problem in the multi-agent system.

Definition 1: We say that a consensus is reached in a structured time-varying synchronous multi-agent system given by (2) if the sequence $x^{(n)}$ converges to the center $C = \left(\frac{1}{m}, \cdots, \frac{1}{m} \right)^T$ of the simplex S^{m-1}.

The following notations are used in this paper. Let $I = \{1,2,\ldots,m\}$ be an index set, $\mathbb{R}$ be a set of real numbers, $\mathbb{R}^m$ be an *m*–dimensional Euclidean space with the standard inner product $(x, y) = \sum_{i=1}^{m} x_i y_i$. Every element of $\mathbb{R}^m$ is a column vector. Let $x = (x_1, \cdots, x_m)^T$ and $e_k = (\delta_{1k}, \cdots, \delta_{mk})^T$, where $k = \overline{1,m}$ be the standard basis of $\mathbb{R}^m$ where δ_{ij} is Kronecker's delta symbol. Let S^{m-1} be an $(m-1)$– dimensional simplex and

$$\operatorname{int} S^{m-1} = \left\{ x \in S^{m-1} :\ x_i > 0,\ \forall i = \overline{1,m} \right\}$$

be its interior. Let $C = \left(\frac{1}{m}, \cdots, \frac{1}{m} \right)^T$ be the center of the simplex S^{m-1}. Let $m(x) = \min_{i=\overline{1,m}} x_i$, $M(x) = \max_{i=\overline{1,m}} x_i$, and $d(x) = M(x) - m(x)$ be functions. Let $\mathbb{P} = (p_{ij})_{i,j=1}^{m}$ be a square matrix and $\mathcal{P} = (P_{i_1 \cdots i_k})_{i_1, \cdots, i_k = 1}^{m, \cdots, m}$ be a *k*–dimensional hypermatrix, where $k \geq 3$ is any natural number.

2. THE MAIN MODEL

In this section, we shall provide some nonlinear protocols of multi-agent systems. We need some preliminary notions and notations.

Definition 2: A *k*–dimensional hypermatrix $\mathcal{P} = (P_{i_1 \cdots i_k})_{i_1, \cdots, i_k = 1}^{m, \cdots, m}$ is called *stochastic* if all its entries are non-negative and it is stochastic in all directions, i.e.,

$$\sum_{i_1=1}^{m} P_{i_1 \cdots i_k} = \cdots = \sum_{i_k=1}^{m} P_{i_1 \cdots i_k} = 1, \quad P_{i_1 \cdots i_k} \geq 0,$$

where for any $i_1, \cdots, i_k = \overline{1,m}$.

Let $\mathcal{P} = (P_{i_1 \cdots i_k})_{i_1, \cdots, i_k = 1}^{m, \cdots, m}$ be a *k*–dimensional stochastic hypermatrix and $\mathcal{P}_l = (P_{i_1 \cdots i_{k-1} l})_{i_1, \cdots, i_{k-1} = 1}^{m, \cdots, m}$ be its $(k-1)$–dimensional l^{th} – subhypermatrix for fixed $l = \overline{1,m}$. It is clear that $\mathcal{P}_l$ is also stochastic for every $l = \overline{1,m}$. In the sequel, we write the dimensional stochastic hypermatrix $\mathcal{P} = \left(\mathcal{P}_1 \middle| \cdots \middle| \mathcal{P}_m \right)$. We define a polynomial operator $V : \mathbb{R}^m \to \mathbb{R}^m$ associated with the *k*–dimensional stochastic hypermatrix $\mathcal{P} = \left(\mathcal{P}_1 \middle| \cdots \middle| \mathcal{P}_m \right)$ as follows:

$$\left(V(x) \right)_l = \sum_{i_1=1}^{m} \cdots \sum_{i_{k-1}=1}^{m} P_{i_1 \cdots i_{k-1} l} x_{i_1} \cdots x_{i_{k-1}}, \quad \forall l = \overline{1,m}. \tag{3}$$

Since $\sum_{l=1}^{m} P_{i_1\cdots i_{k-1}l} = 1$ and $P_{i_1\cdots i_k} \geq 0$, we have that $V(S^{m-1}) \subset S^{m-1}$. In this sense, the polynomial operator $V: S^{m-1} \to S^{m-1}$ defined by (3) is called stochastic.

Let $\mathcal{P}_l = (P_{i_1\cdots i_{k-1}l})_{i_1,\cdots,i_{k-1}=1}^{m,\cdots,m}$ be the lth–subhypermatrix of the k–dimensional stochastic hypermatrix $\mathcal{P} = (P_{i_1\cdots i_k})_{i_1,\cdots,i_k=1}^{m,\cdots,m}$ for fixed $l = \overline{1,m}$.

We also define a polynomial stochastic operator $V_l: S^{m-1} \to S^{m-1}$ associated with the subhypermatrix $\mathcal{P}_l$ as follows:

$$\big(V_l(x)\big)_j = \sum_{i_1=1}^{m} \cdots \sum_{i_{k-2}=1}^{m} P_{i_1\cdots i_{k-2}jl}\, x_{i_1} \cdots x_{i_{k-2}}, \quad \forall j = \overline{1,m}. \tag{4}$$

Due to (3) and (4), it is easy to see that

$$\big(V(x)\big)_l = \big(V_l(x), x\big), \quad \forall l = \overline{1,m}.$$

Therefore, the polynomial stochastic operator $V: S^{m-1} \to S^{m-1}$ given by (3) can be written as follows:

$$V(x) = \Big(\big(V_1(x), x\big), \cdots \big(V_m(x), x\big)\Big)^T, \tag{5}$$

where $V_l: S^{m-1} \to S^{m-1}$ is defined by (4) for $l = \overline{1,m}$.

In the sequel, we write the polynomial stochastic operator $V: S^{m-1} \to S^{m-1}$ as follows $V = \big(V_1 \big| \cdots \big| V_m\big)$.

We now define an square matrix

$$\mathbb{P}_x = \begin{pmatrix} \big(V_1(x)\big)^T \\ \vdots \\ \big(V_m(x)\big)^T \end{pmatrix} = \begin{pmatrix} \big(V_1(x)\big)_1 & \cdots & \big(V_1(x)\big)_m \\ \vdots & \ddots & \vdots \\ \big(V_m(x)\big)_1 & \cdots & \big(V_m(x)\big)_m \end{pmatrix}. \tag{6}$$

It follows from (5) and (6) that

$$V(x) = \mathbb{P}_x x. \tag{7}$$

We call (7) *a matrix form* of the polynomial stochastic operator $V: S^{m-1} \to S^{m-1}$ defined by (3).

We now show that the square matrix $\mathbb{P}_x$ is doubly stochastic for every $x \in S^{m-1}$. Since $V(S^{m-1}) \subset S^{m-1}$ for any $x \in S^{m-1}$, the matrix $\mathbb{P}_x$ is row stochastic. On the other hand, it follows form (4) that

$$\begin{aligned} \sum_{l=1}^{m} \big(V_l(x)\big)_j &= \sum_{l=1}^{m} \left(\sum_{i_1=1}^{m} \cdots \sum_{i_{k-2}=1}^{m} P_{i_1\cdots i_{k-2}jl}\, x_{i_1} \cdots x_{i_{k-2}} \right) \\ &= \sum_{i_1=1}^{m} \cdots \sum_{i_{k-2}=1}^{m} \left(\sum_{l=1}^{m} P_{i_1\cdots i_{k-2}jl} \right) x_{i_1} \cdots x_{i_{k-2}} \\ &= \big(x_1 + \cdots + x_m\big)^{k-2} = 1. \end{aligned}$$

Thus, the polynomial stochastic operator $V: S^{m-1} \to S^{m-1}$ defined by (3) can be written in the matrix form (7) where $\mathbb{P}_x$ is doubly stochastic for every $x \in S^{m-1}$. The polynomial stochastic operator (3) has an incredible application in physics, biology, economics. The simplest nonlinear polynomial stochastic operator is a quadratic stochastic operator associated with a cubic stochastic hypermatrix. The quadratic stochastic operator (in short QSO) has a fascinating application in population genetics [47]. The QSO describes a distribution of the next generation in the population system if the distribution of the current generation was given. In the paper [48], a mathematical model of a transmission of human ABO blood groups was described as the QSO on 7-dimensional simplex and based on some numerical investigations of QSO, the future ABO blood group distribution of Malaysian people was predicted. In the paper [49], it was given a long self-contained exposition of the recent achievements and open problems in the theory of QSO.

The main problem in the nonlinear operator theory is to study the behavior of nonlinear operators. This problem was not fully finished even in the class of QSO (the QSO is the simplest nonlinear operator) (see [49] for more details). In [50,51], a special class of QSO was studied as a generalization of a logistic mapping into the higher dimension.

We figured out that the matrix form (7) of the polynomial stochastic operator defined by (3) gives an advantage during the study of its dynamics. More precisely, on the one hand, the trajectory $\{x^{(n)}\}$ of $V: S^{m-1} \to S^{m-1}$ starting from $x^{(0)} \in S^{m-1}$, where $x^{(n+1)} = V(x^{(n)}) = \mathbb{P}_{x^{(n)}} x^{(n)}$, presents a nonlinear protocol of an opinion sharing dynamics of the multi-agent system. On the other hand, thanks again to the matrix form (7), the trajectory of the polynomial stochastic operator $V: S^{m-1} \to S^{m-1}$ is nothing more than non-homogeneous Markov chains $\{\mathbb{P}_{x^{(n)}}\}$ with some initial distribution $x^{(0)} \in S^{m-1}$ (in our case, we actually have doubly stochastic matrices and it would be a powerful usage). In this paper, by means of the theory of non-homogeneous Markov chains, we shall establish a consensus in the multi-agent system.

Now, we are ready to give our nonlinear protocol of multi-agent systems.

Protocol A: Let $\mathcal{P} = (P_{i_1\cdots i_k})_{i_1,\cdots,i_k=1}^{m,\cdots,m}$ be a k–dimensional stochastic hypermatrix and $V: S^{m-1} \to S^{m-1}$ be an associated polynomial stochastic operator defined by (3). Suppose that an opinion sharing dynamics of the multi-agent system is given as a trajectory of a polynomial stochastic operator

$$x^{(n+1)} = V(x^{(n)}), \quad x^{(0)} \in S^{m-1}, \tag{9}$$

where $x^{(n)} = (x_1^{(n)}, \cdots, x_m^{(n)})^T$ is the probabilistic subjective distribution of the members after n–step revisions.

3. RESULTS AND DISCUSSIONS

It is clear that all functions $m, M, d: S^{m-1} \to \mathbb{R}$ are continuous where $m(x) = \min_{i=\overline{1,m}} x_i$, $M(x) = \max_{i=\overline{1,m}} x_i$, $d(x) = M(x) - m(x)$ for any $x \in S^{m-1}$. Moreover, one has that $d(x) = 0$ if and only if $x = C = \left(\frac{1}{m}, \cdots, \frac{1}{m}\right)^T$.

We need the following simple but crucial lemma.

Lemma 1: A sequence $\{x^{(n)}\} \subset S^{m-1}$ converges to $C = \left(\frac{1}{m}, \cdots, \frac{1}{m}\right)^T$ if and only if $\lim_{n\to\infty} d(x^{(n)}) = 0$.

Proof: Let $\{x^{(n)}\} \subset S^{m-1}$ be any sequence.

The "Only if" part. If sequence $\{x^{(n)}\} \subset S^{m-1}$ converges to $C = \left(\frac{1}{m}, \cdots, \frac{1}{m}\right)^T$ then we have that

$$\lim_{n\to\infty} d\left(x^{(n)}\right) = M(C) - m(C) = 0$$

The "If" part. Suppose that $\lim_{n\to\infty} d(x^{(n)}) = 0$. Let $x^* \in \omega(\{x^{(n)}\}) \neq \varnothing$ be any limiting point. Then there is a subsequence $\{x^{(n_k)}\}$ such that $\lim_{k\to\infty} x^{(n_k)} = x^*$. Then we get that $d(x^*) = \lim_{k\to\infty} d(x^{(n_k)}) = 0$. Therefore, we get that $x^* = C$. This is the end the proof.

Theorem 1: Suppose that an opinion sharing dynamics in the multi-agent system is given by nonlinear Protocol A. If every member of the people has a positive subjective opinion on the given task after some revision steps then the multi-agent system reaches to a consensus.

Proof: The theorem states that if $x^{(n_0)} \in \operatorname{int} S^{m-1}$ for some $n_0 \in \mathbb{N}$ then the sequence defined by (9) converges to the center $C = \left(\frac{1}{m}, \cdots, \frac{1}{m}\right)^T$ of the simplex S^{m-1}. In order to prove it, due to Lemma 1, it is enough to show that $\lim_{n\to\infty} d(x^{(n)}) = 0$. Without loss of any generality, we may suppose that $x^{(0)} \in \operatorname{int} S^{m-1}$.

Due to the matrix form (7) of the polynomial stochastic operator given by (3), the opinion sharing dynamics (9) can be written as $x^{(n+1)} = \mathbb{P}_{x^{(n)}} x^{(n)}$ where $\mathbb{P}_x$ is the square doubly stochastic matrix given by (6).

We want to show that

$$m(x^{(0)}) \le \cdots \le m(x^{(n)}) \le \cdots \le M(x^{(n)}) \le \cdots \le M(x^{(0)}).$$

In fact, since $\mathbb{P}_x = (p_{ij}(x))_{i,j=1}^m$ is the square doubly stochastic matrix, we get that

$$\begin{aligned} M\left(x^{(n+1)}\right) &= x_{i_0}^{(n+1)} = \sum_{j=1}^m p_{i_0 j}\left(x^{(n)}\right) x_j^{(n)} \\ &\le M\left(x^{(n)}\right) \sum_{j=1}^m p_{i_0 j}\left(x^{(n)}\right) = M\left(x^{(n)}\right), \\ m\left(x^{(n+1)}\right) &= x_{i_1}^{(n+1)} = \sum_{j=1}^m p_{i_1 j}\left(x^{(n)}\right) x_j^{(n)} \\ &\ge m\left(x^{(n)}\right) \sum_{j=1}^m p_{i_1 j}\left(x^{(n)}\right) = m\left(x^{(n)}\right). \end{aligned}$$

We now show that $p_{ij}(x^{(n)}) \in [m(x^{(0)}), M(x^{(0)})]$ for any $n \in \mathbb{N}$ and $i, j = \overline{1,m}$.

Indeed, we have that

$$\begin{aligned} p_{ij}\left(x^{(n)}\right) &= \left(V_i\left(x^{(n)}\right)\right)_j = \sum_{i_1=1}^m \cdots \sum_{i_{k-2}=1}^m P_{i_1 \cdots i_{k-2} ji}\, x_{i_1}^{(n)} \cdots x_{i_{k-2}}^{(n)} \\ &\le M\left(x^{(0)}\right) \sum_{i_1=1}^m \cdots \sum_{i_{k-3}=1}^m x_{i_1}^{(n)} \cdots x_{i_{k-3}}^{(n)} \left(\sum_{i_{k-2}=1}^m P_{i_1 \cdots i_{k-2} ji} \right) \\ &= M(x^{(0)}), \\ p_{ij}\left(x^{(n)}\right) &= \left(V_i\left(x^{(n)}\right)\right)_j = \sum_{i_1=1}^m \cdots \sum_{i_{k-2}=1}^m P_{i_1 \cdots i_{k-2} ji}\, x_{i_1}^{(n)} \cdots x_{i_{k-2}}^{(n)} \\ &\ge m\left(x^{(0)}\right) \sum_{i_1=1}^m \cdots \sum_{i_{k-3}=1}^m x_{i_1}^{(n)} \cdots x_{i_{k-3}}^{(n)} \left(\sum_{i_{k-2}=1}^m P_{i_1 \cdots i_{k-2} ji} \right) \\ &= m(x^{(0)}). \end{aligned}$$

We now show that

$$d\left(x^{(n+1)}\right) \le \left(1 - 2m\left(x^{(0)}\right)\right) d\left(x^{(n)}\right).$$

In fact, we have that

$$\begin{aligned} x_i^{(n+1)} &= \sum_{j=1}^m p_{ij}\left(x^{(n)}\right)\left(x_j^{(n)} - M\left(x^{(n)}\right)\right) + M\left(x^{(n)}\right) \\ &\le m\left(x^{(0)}\right)\left(m\left(x^{(n)}\right) - M\left(x^{(n)}\right)\right) + M\left(x^{(n)}\right), \\ x_i^{(n+1)} &= \sum_{j=1}^m p_{ij}\left(x^{(n)}\right)\left(x_j^{(n)} - m\left(x^{(n)}\right)\right) + m\left(x^{(n)}\right) \\ &\ge m\left(x^{(0)}\right)\left(M\left(x^{(n)}\right) - m\left(x^{(n)}\right)\right) + m\left(x^{(n)}\right), \\ d\left(x^{(n+1)}\right) &= M\left(x^{(n+1)}\right) - m\left(x^{(n+1)}\right) \\ &\le \left(1 - 2m\left(x^{(0)}\right)\right)\left(M\left(x^{(n)}\right) - m\left(x^{(n)}\right)\right) \\ &= \left(1 - 2m\left(x^{(0)}\right)\right) d\left(x^{(n)}\right). \end{aligned}$$

Consequently, we have that

$$d\left(x^{(n)}\right) \le \left(1 - 2m\left(x^{(0)}\right)\right)^n d\left(x^{(0)}\right).$$

Since $x^{(0)} \in \operatorname{int} S^{m-1}$, $1 - 2m(x^{(0)}) < 1$, we get that $\lim_{n\to\infty} d(x^{(n)}) = 0$. This is the end the proof.

Corollary 1: Suppose that an opinion sharing dynamics in the multi-agent system is given by nonlinear Protocol A. If all entries $P_{i_1 \cdots i_k} > 0$, $\forall i_1, \cdots, i_k = \overline{1,m}$ of the *k*–dimensional stochastic hypermatrix are positive, then the multi-agent system reaches to a consensus.

Proof: It is easy to check that if all entries $P_{i_1 \cdots i_k} > 0$, $\forall i_1, \cdots, i_k = \overline{1,m}$ of a *k*–dimensional stochastic hypermatrix $\mathcal{P} = (P_{i_1 \cdots i_k})_{i_1, \cdots, i_k = 1}^{m, \cdots, m}$ are positive, then $V(x^{(0)}) = x^{(1)} \in \operatorname{int} S^{m-1}$. Then, thanks to Theorem 1, we reach to a consensus in the multi-agent system.

Remark 1: We know from the theory of Markov chains that if all entries of a doubly stochastic matrix $\mathbb{P}$ are positive then its trajectory $\{x^{(n)}\}$, where $x^{(n+1)} = \mathbb{P}_n x^{(n)}$, starting from any initial point $x^{(0)} \in S^{m-1}$ converges to the center $C = \left(\frac{1}{m}, \cdots, \frac{1}{m}\right)^T$ of the simplex S^{m-1} (i.e., it is regular). The similar result for the *k*–dimensional stochastic hypermatrix $\mathcal{P} = (P_{i_1 \cdots i_k})_{i_1, \cdots, i_k = 1}^{m, \cdots, m}$ was open. It is worth of pointing out that Corollary 1 is the pioneering result for a multidimensional stochastic hypermatrix.

Remark 2: In the Protocol A, we suppose that a dimensional hypermatrix is stochastic (i.e., elements are nonnegative and it is stochastic in all directions). However, in the proof of Theorem 1, we only utilize the following conditions

$$\sum_{i_k=1}^{m} P_{i_1\cdots i_k} = \sum_{i_{k-1}=1}^{m} P_{i_1\cdots i_k} = \sum_{i_{k-2}=1}^{m} P_{i_1\cdots i_k} = 1 \quad P_{i_1\cdots i_k} \geq 0.$$

Therefore, our all results remain to be true if we only require that a multidimensional hypermatrix having nonnegative entries is stochastic only in three directions (without loss of generality, we may assume that the last direction is one of these three directions).

4. NUMERICAL RESULTS

In this section, we shall numerically investigate some concrete models. Let us consider the following cubic triple stochastic matrix $\mathcal{P} = (\mathcal{P}_1 | \mathcal{P}_2 | \mathcal{P}_3)$ where $\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3$ are square doubly stochastic matrices given as follows

$$\mathcal{P}_1 = (a \quad b \quad c), \ \mathcal{P}_2 = (b \quad c \quad a), \ \mathcal{P}_3 = (c \quad a \quad b),$$

where

$$a = (a_1, a_2, a_3)^T, \quad b = (b_1, b_2, b_3)^T, \quad c = (c_1, c_2, c_3)^T$$

are column vectors of matrices such that

$$a_1 + a_2 + a_3 = b_1 + b_2 + b_3 = c_1 + c_2 + c_3 = 1,$$
$$a_1 + b_1 + c_1 = a_2 + b_2 + c_2 = a_3 + b_3 + c_3 = 1,$$
$$a_1,\ a_2,\ a_3,\ b_1,\ b_2,\ b_3,\ c_1,\ c_2,\ c_3 \geq 0.$$

We define $\mathbb{P}_x = x_1\mathcal{P}_1 + x_2\mathcal{P}_2 + x_3\mathcal{P}_3$ and

$$V(x) = \mathbb{P}_x x.$$

Let us consider the following examples.

Case 1:

$a_1 = 0.2$ $a_2 = 0.5$ $a_3 = 0.3$
$b_1 = 0.7$ $b_2 = 0.15$ $b_3 = 0.15$
$c_1 = 0.1$ $c_2 = 0.35$ $c_3 = 0.55$
$x_1^{(0)} = 0.3$ $x_2^{(0)} = 0.5$ $x_3^{(0)} = 0.2$

Case 2:

$a_1 = 0.75$ $a_2 = 0.15$ $a_3 = 0.1$
$b_1 = 0.2$ $b_2 = 0.5$ $b_3 = 0.3$
$c_1 = 0.05$ $c_2 = 0.35$ $c_3 = 0.6$
$x_1^{(0)} = 0.1$ $x_2^{(0)} = 0.2$ $x_3^{(0)} = 0.7$

All these numerical examples show that the multi-agent system eventually reaches to a consensus as fast as geometric progression.

5. CONCLUSION

The novelty of the paper is that a new nonlinear protocol was presented in a structured time-varying and synchronous multi-agent system as a generalization of the classical models of DeGroot and Chatterjee-Seneta. The opinion sharing dynamics in the multi-agent system is given by a trajectory of a polynomial stochastic operator associated with a multidimensional stochastic hypermatrix (Protocol A). We showed that the multi-agent system eventually reaches to a consensus if either of the following two conditions is satisfied: (i) every member of the people has a positive subjective opinion on the given task after some revision steps or (ii) all entries of the multidimensional stochastic hypermatrix are positive. Numerical results are also presented.

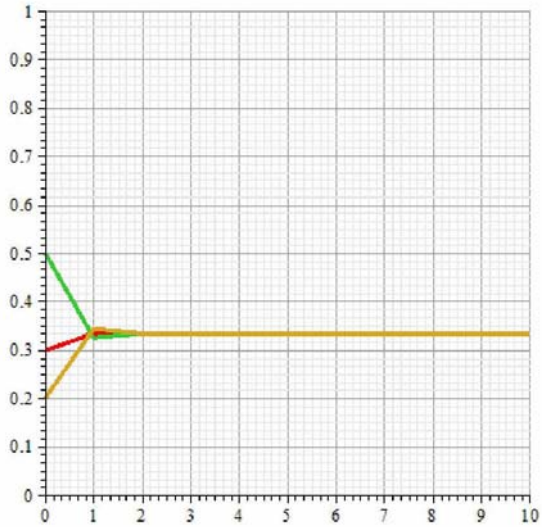


Fig. 1. The simulation of Case 1.

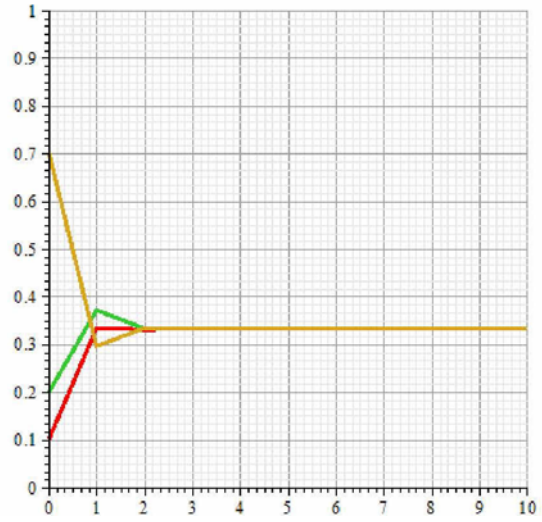


Fig. 2. The simulation of Case 2.

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